

BLAIR, NE, USA

WMO: 720405

Lat: 41.412N

Lon: 96.109W

Elev: 1299

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00436

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.3	1.9	-10.9	3.2	-0.8	-6.3	4.1	4.0	30.1	22.8	27.5	24.0	7.6	320	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	93.0	74.8	90.3	74.3	87.8	73.7	78.7	88.0	77.0	86.2	75.3	84.8	12.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.0	142.4	85.5	73.7	131.8	83.2	72.3	125.7	81.9	43.3	88.1	41.2	86.4	39.7	84.6	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.3	23.6	20.4	DB	-10.6	97.7	4.8	2.9	-14.0	99.8	-16.8	101.4	-19.5	103.1	-23.0	105.2	(4)
			WB	-11.1	81.5	4.7	1.4	-14.5	82.5	-17.2	83.3	-19.8	84.1	-23.2	85.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	24.2	26.8	41.3	51.6	63.0	73.3	76.1	73.7	67.5	53.7	40.5	27.9	(6)
		DBStd	20.83	12.05	12.04	12.29	10.07	9.37	6.16	5.77	5.23	7.71	9.31	10.75	11.38	(7)
		HDD50	2964	799	654	326	102	11	0	0	1	69	317	685		(8)
		HDD65	5968	1264	1071	740	413	153	7	2	62	366	736	1150		(9)
		CDD50	3601	0	3	55	150	413	700	808	733	525	182	31	1	(10)
		CDD65	1131	0	0	4	11	90	257	346	272	136	15	0	0	(11)
		CDH74	10210	0	0	48	139	847	2411	3332	2194	1110	124	5	0	(12)
		CDH80	3654	0	0	9	36	297	906	1297	741	344	24	0	0	(13)
(14)	Wind	WSAvg	9.6	10.1	10.4	11.4	12.1	10.4	9.1	7.2	7.1	8.3	9.3	10.5	9.9	(14)
(15)	Precipitation	PrecAvg	30.3	0.6	0.6	2.0	3.0	4.1	4.0	3.8	3.3	3.6	2.5	1.3	0.9	(15)
(16)		PrecMax	41.7	1.6	2.4	5.3	7.1	9.0	8.8	9.6	7.9	14.1	5.4	5.1	4.4	(16)
(17)		PrecMin	21.3	0.0	0.0	0.0	0.4	0.6	0.0	0.0	0.6	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	5.9	0.5	0.5	1.7	1.8	1.9	2.3	2.2	2.2	2.6	1.6	1.1	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	55.7	65.2	79.4	84.9	93.4	95.8	97.1	95.0	91.7	82.1	72.9	58.3		(19)
(20)		0.4% MCWB	43.5	52.2	59.3	62.9	70.7	74.0	75.8	72.6	73.4	63.2	56.5	49.4		(20)
(21)		2% DB	49.2	54.8	72.4	77.6	87.5	91.3	92.8	90.4	88.2	77.2	66.2	52.3		(21)
(22)		2% MCWB	41.9	45.1	55.8	58.8	67.4	74.0	76.7	74.8	72.7	60.8	54.1	45.8		(22)
(23)		5% DB	44.3	48.9	66.0	72.6	82.4	88.5	90.2	87.2	84.1	72.9	61.1	47.9		(23)
(24)		5% MCWB	38.2	42.4	54.0	56.7	65.0	72.9	76.7	74.4	71.0	58.4	51.1	41.7		(24)
(25)		10% DB	39.9	43.2	60.6	68.1	78.7	85.4	86.9	84.0	80.7	68.9	55.7	43.0		(25)
(26)		10% MCWB	35.8	38.3	50.6	54.9	63.3	71.2	75.4	72.4	68.3	57.1	47.1	38.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	45.4	53.7	63.3	65.4	72.7	79.4	81.5	80.2	76.4	68.3	59.5	53.1		(27)
(28)		0.4% MCDB	54.0	65.6	72.2	77.7	83.7	89.9	91.3	89.6	87.0	75.3	67.3	55.3		(28)
(29)		2% WB	41.7	47.0	59.6	62.1	70.4	77.0	79.4	77.4	74.6	64.7	55.3	46.0		(29)
(30)		2% MCDB	47.6	54.2	68.5	73.6	82.4	87.2	88.9	85.7	84.2	73.5	64.9	50.0		(30)
(31)		5% WB	39.0	42.1	54.9	59.5	68.1	75.1	77.7	75.6	73.0	61.9	51.9	41.9		(31)
(32)		5% MCDB	43.6	48.4	64.7	69.8	79.3	86.0	86.8	84.2	81.5	69.1	59.9	47.6		(32)
(33)		10% WB	36.1	38.7	50.4	56.9	65.8	73.2	75.9	73.8	71.2	58.7	48.1	38.7		(33)
(34)		10% MCDB	40.1	43.6	61.6	66.0	75.9	83.0	85.1	82.0	78.2	66.8	55.9	43.1		(34)
(35)	Mean Daily Temperature Range	MDBR	16.6	16.6	19.2	20.1	19.5	18.4	17.6	17.7	19.7	19.8	18.2	15.2		(35)
(36)		5% DB MCDBR	22.4	23.9	29.6	27.7	25.0	21.2	19.2	21.0	22.7	25.6	24.7	22.2		(36)
(37)		5% DB MCWBR	15.9	15.7	15.8	14.2	10.1	9.8	9.5	9.7	10.3	13.0	14.2	15.7		(37)
(38)		5% WB MCDBR	20.0	22.0	24.8	22.6	21.4	18.8	17.3	18.3	18.7	21.1	22.6	20.5		(38)
(39)		5% WB MCWBR	14.4	15.2	14.5	13.6	10.1	10.6	10.2	9.8	9.9	13.0	14.2	15.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.280	0.313	0.329	0.387	0.411	0.415	0.425	0.415	0.375	0.330	0.296	0.283		(40)
(41)		taud	2.445	2.376	2.396	2.246	2.251	2.298	2.266	2.298	2.368	2.484	2.504	2.455		(41)
(42)		Ebn at Noon	279	284	292	280	275	273	269	268	272	275	272	268		(42)
(43)		Edh at Noon	24	30	34	42	43	41	42	40	34	27	23	22		(43)
(44)	All-Sky Solar Radiation	RadAvg	631	933	1206	1549	1837	2043	2062	1774	1462	1021	711	531		(44)
(45)		RadStd	59	55	109	111	137	158	133	120	94	120	48	53		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page